
The UFO Phenomenon: Laugh, Laugh, Study, Study

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by J. Allen Hynek

Those who are not shocked when they first come across quantum mechanics cannot possibly have understood it. -- Niels Bohr.

Quantum physics arose from the fires of paradox. Its concepts were shocking to those trained in classical physics, and we needed time to feel comfortable with them. As in other areas of science, observations that initially didn't "fit" ultimately led to breakthroughs, while those that did conform to the current scientific paradigm yielded the next decimal places.

The phenomenon of "unidentified flying objects" (UFOs), though few would call it part of science, is also both shocking and paradoxical. UFO reports are often an outrage to common sense. The persistent flow of reports, from all parts of the world, of an object or a luminosity in the sky or on or near the earth defies rational explanation even when made by persons of acknowledged responsibility. (I exclude here the popular *interpretations* of the phenomenon -- such as "little green men from outer space" -- as quite distinct.) Indeed, if these observations are even partially true, understanding them may require a breakthrough -- at least in our thinking about the world around us.

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A Persistent Craze

I first became involved with UFO reports in 1948 -- I was then an astronomer at Ohio State University -- when the Air Force's Air Technical Intelligence Center asked me to help to determine how many of the current sightings had possible astronomical origins as meteors, planets, and twinkling stars. I was squarely in the ranks of those who were sure that the reporting of flying saucers (as they were called) was simply a postwar craze that, like all fads, would quickly run its course. Yet UFO reports have proven to be at least a long-lived "craze"; three decades later it persists, in many levels of society, and in many areas of the world.

The comprehensive catalogue of UFO reports maintained at the Center for UFO Studies contains entries from some 140 countries. Not only is the global ubiquity of the UFO phenomenon undeniable, but the same sorts of sightings are reported from diverse cultures, climates, and levels of sophistication. There appears to be a high awareness of the concept - A Gallup poll has amply verified this in the United States - and every major language has an appropriate term for UFOs. Moreover, critics who hold that interest in UFOs is largely generated by the media may be surprised to learn that sightings have been reported in countries in which discussion of UFOs, especially by the media, is sternly discouraged; the Soviet Union and China are cases in point.

The most useful reports come from people who are sophisticated, responsible, and mentally stable (as judged by commonly accepted standards), if only because they have so much to lose by "going public." Their

"experiences" are almost certain to be greeted with disbelief, even ridicule, by their colleagues. Consider these examples:

- Some years ago an M.I.T. professor called me from across the country. "Drop everything and come to Cambridge," he said, to investigate a sighting by one of his colleagues, the associate director of the Instrumentation Laboratory. I did so and listened with interest to his technical description of the trajectory and appearance (even as to its color-temperature) of a light coursing through the sky. Its origin remains unidentified.
- I recently interviewed a high-ranking officer of the Argentine Air Force, in the presence of other officers, who had sighted a UFO some months before. He was driving through open country late one afternoon with a colleague, in threatening weather, when a cone-shaped object suddenly appeared out of low-hanging clouds and proceeded to move at a moderate speed just beneath the cloud cover. He immediately stopped the car and had only enough time to step out and snap one photograph before the object rose back into the clouds. I examined the original negative under magnification; it appeared to be a bona fide image, with no evidence of darkroom trickery.
- A four-person crew from the Army Medical Rescue Mission was flying a helicopter to Cleveland from Columbus, Ohio when they encountered a wingless craft that swooped toward them, hovered momentarily, and seriously affected the behavior of the helicopter: a strange force reportedly raised it while the pilot was attempting an emergency descent.
- A commercial pilot - a commandante for many years with Alitalia - was flying at 12,000 feet on a sunny day when he and his crew observed a strange "metallic" craft flying alongside. It paralleled the airplane for a few moments, then suddenly rose vertically and disappeared into the sky. Upon reporting a "strange balloon" (he was not going to report a UFO), the pilot was told that the crew of a British airliner at 40,000 feet, an instant before, had also reported an "unknown" object swishing by on a vertical trajectory into the sky. But the pilot did not report the incident to his company, knowing full well that airlines do not look favorably upon those who "see things."

As technically trained as some of the UFO witnesses may be, we must face the fact that most individual UFO reports are anecdotal. Lack of support for professional investigation has undoubtedly let many opportunities to obtain scientific data slip through the cracks. Nonetheless, some investigators, largely in their spare time, have succeeded in relating UFO reports to physical parameters. C. Poher was able to show a statistical correlation between UFO in France and the vertical component of the geomagnetic field as recorded at the Chambon-La-Forêt Geophysical Station. More recently, J. Accetta (under a grant from the Center of UFO Studies) conducted a search for perturbations in routinely recorded geophysical data (housed at the World Data Center and maintained by the U.S. National Oceanic and Atmospheric Administration) that might corroborate UFO sightings. Of some 65 categories of geophysical data, including solar, interplanetary, ionospheric, and geomagnetic phenomena, he found six categories that seemed to merit further attention. D. Pearson has described in some detail a system for retrospective measurement of ground traces allegedly associated with UFO sightings that includes several thermal-analysis methods for determining to what extent the soil at the UFO site had been heated. But lack of funds has hampered such investigations, although very promising (unpublished) explorations have occurred.

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UFO researchers are caught in a classic "Cath-22" situation: when they seek funds and projects to collect quantitative evidence, critics point to the absence of supporting data. For example, NASA, in rejecting

President Carter's request that it undertake an examination of the UFO question, stated in effect that it would do so only if presented with "hard" data. Some data-collecting systems exist for other purposes, but the lack of legitimacy makes it virtually impossible for UFO researchers to exploit them. For example, the North American Air Defense Command could introduce a subroutine in its computer program to monitor the many UCTs (uncorrelated targets) it daily observes on radar, but when I suggested this to the Air Force I was told that their mission is to check only ballistic trajectories.

Beyond the Fringe

UFOs are difficult to take seriously, and much of the derision from the scientific community is well deserved. Three aspects in particular have led to their general dismissal: the preponderance of *identified* flying objects (IFOs); the space-age spawned belief in the concept of "we are not alone" (and its corollary, "We've gone to the moon so why can't they come here?"); and the few but highly visible "true believers" who have adopted the idea of celestial visitors with quasi-religious fervor.

It is true that the great majority of initial UFO reports are simply the result of misidentifications of ordinary events. A Hendry's analysis of the reports received at the Center for UFO Studies over two years showed that nearly 90 percent were identifiable. Clearly, if that many people can be mistaken, why not assume that *all* reports are either misidentifications or hoaxes? But such dismissal does not resonate with the scientific outlook. After all, only one unexplained track in a helium bubble chamber out of thousands indicates a new subatomic particle.

Stars have twinkled, planets have risen and set, and meteors have flashed across the skies for untold centuries; why are they now suddenly being reported as UFOs? Perhaps the answer lies in the tenor of the times, which is really up to sociologists and psychologists to explain. In this age of unyielding tension from the spectre of nuclear holocaust, dwindling natural resources, overpopulation, pollution, inflation, the energy crisis, and the breakdown of social traditions, wouldn't it be nice if we could put all our troubles on someone else's shoulders? But whose? Why, the extraterrestrials, of course! Throughout history people have looked at the skies for succor, but the space age has replaced the gods and spirits of old with the enticing possibility that intelligence far more advanced than our own is visiting the earth. After all, if they can get here, they must have very advanced technology, and we could be the beneficiaries of their fabulous knowledge.

From this conviction it may be just a short step to misidentifying what one sees in the sky - or to seeing UFOs simply because one wants to see them. And there are always those small but colorful "space people" cults that blindly accept their leaders' accounts of trips to Venus on UFOs and the lofty messages relayed to humanity from those who make their home there.

There emotional, even neurotic aspects of the UFO scene could easily lead to the conclusion that the UFO phenomenon is utter rubbish. But this impugns the integrity, and perhaps the competence, of our scientists, pilots, engineers, and others judged sane and responsible who have related sober albeit incredible accounts of UFO encounters. These certainly cannot be put into the same category as alleged visits to Venus and Mars.

Call in the Professionals

After many years of experience with virtually all aspects of the UFO phenomenon, I have come to believe that if we "precipitate out" the essential elements from the chaos of "popular ufology," we will uncover a new empirical phenomenon, perhaps comparable to the first glimpses of microorganisms by Leeuwenhoek or Jupiter's satellites by Galileo. Unfortunately, the process may be almost as taxing as Madame Curie's extraction of a gram of radium from several tons of pitchblende.

This hasn't already been done because in the face of overwhelming ridicule, it has been impossible to obtain qualified personnel and the necessary funds to treat the subject seriously and professionally. In the wake of buffoonery and religious fantasy, the field has been left to the well-meaning but untrained amateur who all too often has fallen into the same trap as the scientist - of equating the UFO phenomenon irrevocably with "SETI" (the search for extraterrestrial intelligence), leaving no room for open-ended research. However, these same amateurs have done yeoman service in gathering and preserving data that otherwise would have been irretrievably lost, and they did this while earning their livelihood elsewhere. What chance would medical research or going to the moon have if left entirely to unpaid volunteers?

Well-known previous efforts - Project Blue Book, the Robertson panel, and the Condon Committee - constituted "professional treatment" only in a very limited sense. As a consultant to Blue Book (the Air Force's UFO project from 1952 to 1969), an associate member of the Robertson Panel, and one who kept in close touch with the activities of the Condon Committee, I can speak with some competence.

Project Blue Book took its signals from the Pentagon and these, largely dictated by civilian and military scientific advisors, were that rational explanations for all UFO reports should be found. But these explanations were rarely based on extended investigations because of circular reasoning: since the great majority of UFO reports can be explained rationally, then if one tries harder almost *all* reports could be so explained; therefore, why bother? Little effort was made to obtain quantitative data - charts, graphs, angular velocities, subtended angles, spectral characteristics, and so on - since UFOs had to be nonsense.

The Robertson Panel, composed of high-ranking (and very busy) scientists, spent parts of five days early in 1953 surveying the situation. It made no investigations of its own, relying solely on fragmentary examinations of cases selected by Blue Book personnel. The panel had been convened by the CIA, whose concern seemed not to be the UFOs per se but the possible use of UFO reports by subversive elements to clog military communications or affect the psychological stability of the public. Instead of suggesting further scientific investigation, the panel recommended that every effort be made to "play down" UFO reports.

There have been many criticisms of the Condon Committee, although its report received the imprimatur of the National Academy of Sciences and has been accepted by some as the definite work on UFOs. A quotation from just one critic - perhaps the mildest - will suffice. The subcommittee on UFOs of the American Institute of Aeronautics and Astronautics (AIAA) stated: "To understand the Condon report, which is difficult to read, due in part to its organization, one must study the bulk of the report. It is not enough to read summaries, such as those by Sullivan and Condon, or summaries of summaries, on which the vast majority of readers and news media seem to rely. There are differences in the opinions and conclusions drawn by the authors of the various chapters, and there are differences between these and Condon's summary." The AIAA group further remarked: "[We were] greatly perturbed by the paucity of thorough scientific and technological analysis applied to practically all observations before the Condon study."

UFO researchers are caught in a classic "Catch-22" situation: when they seek funds for collecting solid evidence, critics point to the absence of solid evidence.

Of course, one could pursue the theory that Project Blue Book and the Condon Committee were part of a super whitewash - that the highest echelons of government, not only of this nation but of many nations, know what is happening but are intentionally covering it up. And I continue to receive clandestine reports from military personnel that they have been involved intimately or peripherally, in such a cover-up but who

plead fear of reprisal when I request a signed statement. Yet even though use of the Freedom of Information Act recently revealed that the CIA and FBI had exhibited interest in UFOS - they stoutly denied it earlier - this hardly constitutes evidence of a sinister, Machiavellian plot. If such a global cover-up indeed exists, it would constitute the best-kept secret of all time.

Putting Reports in Order

If, in due course, grants for professional study of the UFO phenomenon do become available, how might we proceed? The late astronomer Henry Norris Russel set a fine example in *The Origin of the Solar System*. He didn't quite solve the problem, but he did set down the known properties of the solar system (coplanar orbits, the revolution, rotation, masses, and densities of planets and satellites) for which any viable theory must account. We can hardly do better than to follow this example with respect to the UFO phenomenon.

As a first step, we can order reports into six observational categories. These in no way presuppose the origin of the phenomenon; they simply specify the type of UFO experience. The first three categories are observations at a distance, while the last three are "close encounters," close enough for detailed features to be observable (a distance of 200 yards or less is a rule of thumb):

Nocturnal Lights. The Witness observe a luminous point or extended source; in the latter case, the luminosity generally obscures any presumed material form of the source. it might be described as a concentrated source of electromagnetic energy, strong but not necessarily peaking in the visible spectrum.

Daylight Discs. The operative word here is "daylight"; however, since the great majority of sightings made in the daytime refer to discoidal or oval (sometimes cylindrical) metallic-looking objects, I refer to them generally as discs. Whether a nocturnal light would appear as a metallic disc by day is not known. In this purely observational classification system, the classes may or may not overlap. Indeed, we must not assume that all UFOs have the same origin; we may have "apples and oranges."

Radar and Radar-Visual. Radar is the primary source of information, but particularly important are cases in which the UFO has also been sighted visually and the two observations substantially agree.

Close Encounters of the First Kind. There is no reported interaction between the UFO and the environment - These reports are the most common.

Close Encounters of the Second Kind. There is interaction with either or both animate and inanimate matter. The literature is replete with cases in which car engines have been killed at the approach of a UFO, holes and burnt rings on the ground have been found at the exact site of the alleged landing, and physiological effects on people and animals, as well as disturbances or destruction of vegetation, have been reported. These encounters obviously have the most immediate scientific value since they are capable of being studied in the laboratory.

Close Encounters of the Third Kind. These are distinguished by reports of creatures or entities closely associated with the UFO, regardless of whether they interact with human observers. Though the least frequently reported, this category has received by far the greatest prominence in the media because of its obvious appeal to the imagination and the "we are not alone" concept.

Numerous examples of all six categories are available in the literature.

Paradoxical Dualism

In our search for the properties of the UFO phenomenon, do we find anything that sets it apart from the everyday world? Is there something that makes it both shocking and paradoxical in the Niels Bohr sense, and hence that it might suggest where to look for a breakthrough? The answer appears to be yes.

The UFO phenomenon, whatever its origin, is largely localized in both space and time. For example, unlike commercial aircraft, which can be tracked and viewed sequentially as they pass over town after town, a UFO is rarely observed in more than one locality, and virtually never it is seen sequentially. Like the Cheshire Cat in *Alice in Wonderland*, it appears almost out of nowhere, makes its presence known, and then disappears. Also like the cat, it is sometimes reported to "materialize" and "dematerialize" and to change form. Alice's cat left its grin when it departed; the UFO, except in "close encounters of the second kind," leaves only a haunting memory. And like the cat, a UFO's appearance is short-lived - several studies have shown that the duration of a sighting averages 8 to 14 minutes.

"Close encounters of the third kind" are the least frequently reported, yet this category of observations has received by far the greatest prominence in the media.

Alice's cat had only one witness. Records show that about two-thirds of the cases involve two or more witnesses, but they rarely have a host of witnesses. This has been the primary objection of some who might otherwise take UFO reports seriously: why so few witnesses? If we were dealing with a "nuts and bolts" craft launched from some cosmic Cape Canaveral, shouldn't it be visible to a great many earth dwellers?

The selective appearance of the UFO suggests deliberate staging, but on whose part? By whatever intelligence lies behind the UFO phenomenon or by an unconscious effort of the witness? For example, it is frequently stated that a UFO landed "on the road, directly ahead of our car." Why not far off to the side? Why directly in plain view, but then only to a handful of people?

Beyond these reported properties must be added even more bizarre "paranormal" characteristics. In addition to "materialization," dematerialization," and change of form, implausible accelerations, speeds, and "instantaneous" changes in position without any apparent travel time have also been frequently reported. Although seemingly incredible, these paranormal aspects are too well documented to be disregarded.

We are apparently faced with a dualism similar to the wave-particle dualism of light that physicists had to confront a century ago. On the one hand, the UFO exhibits a physical nature: it can be seen and photographed, registers on radar, and it can interact with the environment. On the other hand, it behaves as though it were obeying laws yet unknown to physics. We have a situation that is both shocking and paradoxical - one cannot disregard one aspect in favor of the other just because it doesn't fit.

We may have to accept the possibility that the UFO phenomenon is beyond conventional, straightforward explanation, perhaps as the true source of the sun's light was beyond Kelvin and Helmholtz, who held stoutly to their "contradiction theory" at the close of the nineteenth century - that as the sun shrank under the influence of gravity, potential energy was transformed into kinetic energy. The concept of the sun as a "nuclear energy device" was, of course, totally beyond them. Indeed, when told that fossils from the distant past proved that the sun must have been shining then as at present, Kelvin would have none of it. He told a geologist that he would "give them 10 million years and not a day longer" for the age of the sun. Perhaps if Kelvin had been more of a philosopher, he might have pondered whether the fossils were telling him something. Likewise, perhaps we should ponder whether the UFO phenomenon is telling *us* something.

Mind and Matter

The UFO phenomenon is experienced largely through human consciousness and the human psyche. Laboratory physics attempts to work with "objective reality," but suppose there exists a class of phenomena in which subjective variables enter in the first order? How do we handle their study?

Eugene Wigner, the noted Princeton physicist, wrote that "the present laws of physics are at least incomplete without a translation into terms of mental phenomena. More likely, they are inaccurate, the inaccuracy increasing with the role that life plays in the phenomena considered ... as we consider situations in which consciousness is more and more relevant, the necessity for modifications of the regularities obtained for inanimate objects will be more and more apparent."

It is becoming increasingly apparent to those who study the UFO phenomenon that some modification in approach and methodology is necessary. Do events in the mind represent interlopers from a parallel reality? Or, indeed, are they themselves such parallel realities? Should we look to distant star systems for the solution to UFOs or much closer to a metaterrestrial rather than an extraterrestrial hypothesis?

The paranormal or "psychic" aspects of the UFO phenomenon have generally been taken as sufficient reason for dismissing the entire subject, but such dismissal smacks of scientific irresponsibility. Erwin Schroedinger wrote: "A scientist should be curious and eager to find out." I would hold that we have accumulated enough UFO data over the past three decades to be truly curious about it.

We may have to accept the possibility that the UFO phenomenon is beyond conventional, straightforward explanation: it is a situation both shocking and paradoxical.

To Explain the Inexplicable

There is indeed a growing, although still far from overwhelming, interest in the intriguing mystery of the UFO phenomenon - it just will not dry up and blow away as most of us once expected. In 1976, P.A. Sturrock surveyed the membership of the American Astronomical Society, asking whether the UFO phenomenon deserved scientific study. Among the 1,356 respondents, 23 percent replied "certainly," 30 percent "probably," and 27 percent "possibly" - a total of 80 percent at least mildly in favor. Seven respondents stated that they were actively studying the problem. Surprisingly, (perhaps only to those unfamiliar with the UFO scene), 62 respondents stated that they had witnessed, or had obtained recorded evidence of, and even they could not identify and that they thought might be related to the UFO phenomenon.

Another recent example of scientific interest comes from the USSR Academy of Science. Preferring the term "anomalous atmospheric phenomena," Gindilis, Men'kov, and Petrovskaya report that "the substantial percentage of observers who have adequate qualifications attracts attention: scientific workers, engineers, pilots (52 percent). Contrary to the widespread fallacy, there is a highly significant percentage of astronomers among the observers (7.5 percent)."

Finally, attention should be called to GEPAN (Groupement pour Etudier les Phenomenes Aerospatiaux Nonidentifies), a government-supported scientific team within the French space agency CNES (Centre

Nationale des Etudes Spatiales) that is systematically studying the UFO phenomenon. France is the only country to have officially undertaken such a project. (Characteristically, perhaps, only *French* UFOs are studied.)

When what was once believed to be a passing craze has instead proved persistent, provocative, and vexing, we may well heed the words of the astronomer Pierre Simon Laplace two centuries ago: "We are so far from knowing all the forces of nature and the various modes of their action that it is not worthy of a philosopher to deny phenomena only because they are inexplicable in the present state of our knowledge. The harder it is to acknowledge the existence of phenomena, the more we are obligated to investigate them with increasing care."

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